

Problem 12.1

edge 1

node	1	2	3	4
1	-1	1	0	0
2	0	-1	1	0
3	0	0	1	-1
4	-1	0	0	1

$$(a, a, a, a)$$

$(1, 0, 0, 0)$ is not orthogonal to the null space.

Problem 12.2

$$A^T C A = \begin{bmatrix} 2 & -1 & 0 & -1 \\ -1 & 3 & -2 & 0 \\ 0 & -2 & 4 & -2 \\ -1 & 0 & -2 & 3 \end{bmatrix}$$

$$x = \begin{bmatrix} 3/4 \\ 1/4 \\ 0 \\ 1/4 \end{bmatrix}$$

$$y = \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix}$$